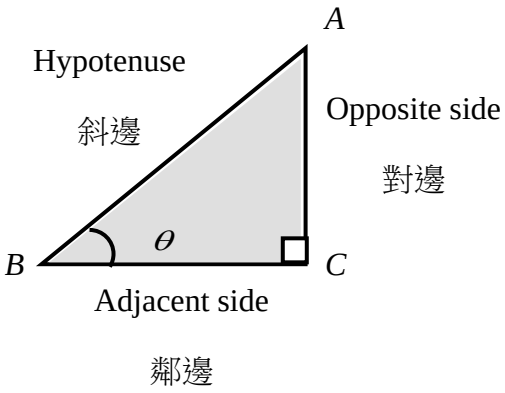
Trigonometry（三角學）

1. Basic Trigonometric Ratio 基本三角比

$$\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}}$$



2. New Trigonometric Ratio 新三角比

$$\text{cosec}\theta = \frac{1}{\sin\theta} \qquad \sec\theta = \frac{1}{\cos\theta} \qquad \cot\theta = \frac{1}{\tan\theta}$$

cosec: cosecant　　sec: secant　　cot : cotangent

(cosecθ 亦可寫成 cscθ.)

Note: 由於計算機沒有 cosec, sec, 及 cot 的按鍵，你必須使用 sin, cos 及 tan 的按鍵然後使用倒數 (1/*x*) 按鍵:

e.g.

$$\text{cosec}30° = \frac{1}{\sin 30°} = 2 \qquad \sec 30° = \frac{1}{\cos 30°} = \frac{2}{\sqrt{3}} \qquad \cot 30° = \frac{1}{\tan 30°} = \sqrt{3}$$

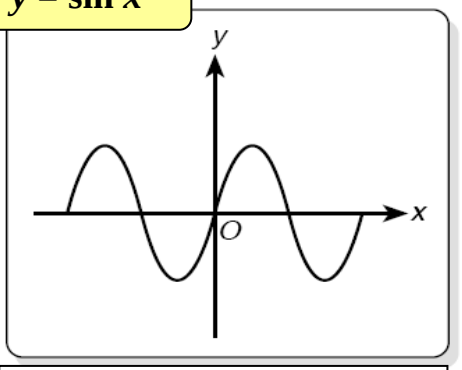
$$\text{cosec}45° = \frac{1}{\sin 45°} = \sqrt{2} \qquad \sec 45° = \frac{1}{\cos 45°} = \sqrt{2} \qquad \cot 45° = \frac{1}{\tan 45°} = 1$$

$$\text{cosec}60° = \frac{1}{\sin 60°} = \frac{2}{\sqrt{3}} \qquad \sec 60° = \frac{1}{\cos 60°} = 2 \qquad \cot 60° = \frac{1}{\tan 60°} = \frac{1}{\sqrt{3}}$$

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Graph of Trigonometric Ratio（三角函數的圖像）

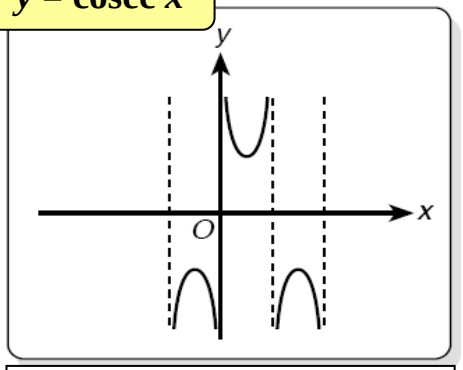
$y = \sin x$



Period (週期) is 360°

$-1 \le \sin x \le 1$

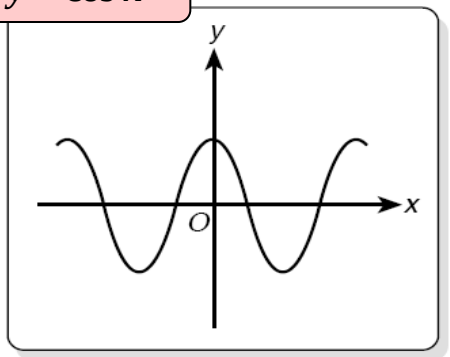
$y = \operatorname{cosec} x$



Period is 360°

$\operatorname{cosec} x \ge 1$ or $\operatorname{cosec} x \le -1$

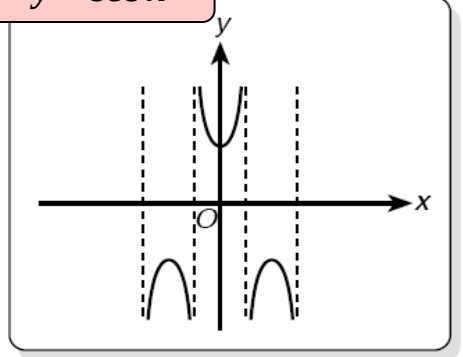
$y = \cos x$



Period is 360°

$-1 \le \cos x \le 1$

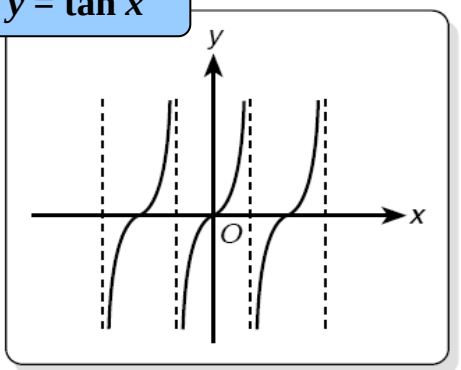
$y = \sec x$



Period is 360°

$\sec x \ge 1$ or $\sec x \le -1$

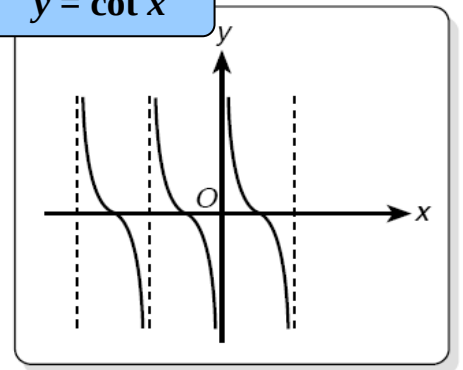
$y = \tan x$



Period is 180°

$\tan x$ is unbounded for all x

$y = \cot x$



Period is 180°

$\cot x$ is unbounded for all x

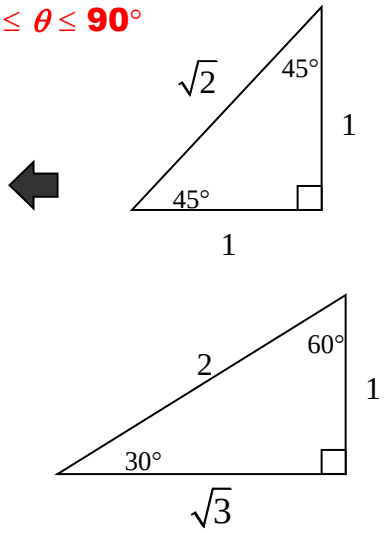
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Trigonometric Ratio（三角比）

1. Trigonometric Ratio of Special Angles $0° \le \theta \le 90°$

特別角度之三角比 **($0° \le \theta \le 90°$)**

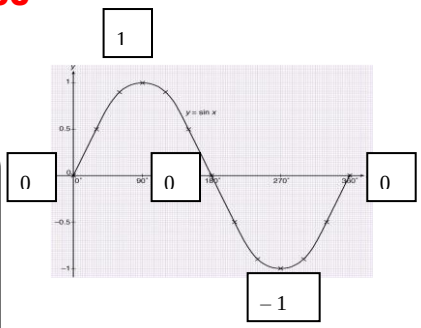
	30°	45°	60°
$\sin\theta$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
$\cos\theta$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
$\tan\theta$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

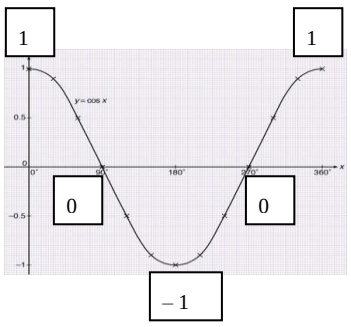


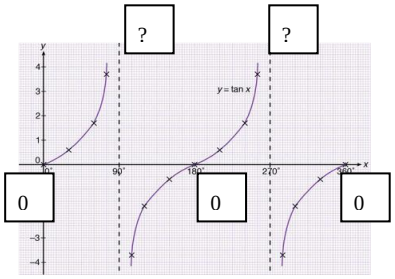
2. Trigonometric Ratio of Special Angles $0° \le \theta \le 360°$

特別角度之三角比 **($0° \le \theta \le 360°$)**

	0°	90°	180°	270°	360°
$\sin\theta$	0	1	0	-1	0
$\operatorname{cosec}\theta$	undefined	1	undefined	-1	undefined
$\cos\theta$	1	0	-1	0	1
$\sec\theta$	1	undefined	-1	undefined	1
$\tan\theta$	0	undefined	0	undefined	0
$\cot\theta$	undefined	0	undefined	0	undefined





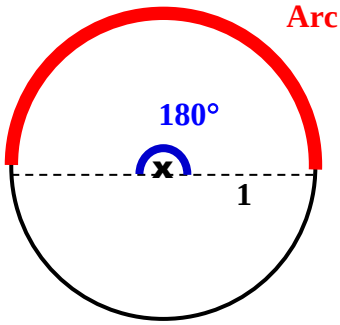


undefined : 不能確定 / 未下定義

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Introduction to Radian（弧度簡介）

1. Measurement of Angles in Radian （以弧度爲單位量度角度）



In Degree 角度		In Radian 弧度
180°		π

Note: 以弧度來量度角度是十分有用的，尤其是在微積分 (Differentiation and Integration).

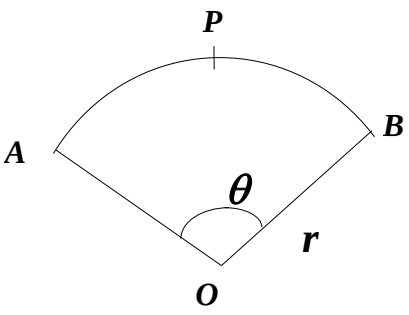
2. Conversion between Degree and Radian （角度與弧度之轉換）

In Degree 角度		In Radian 弧度
1°		$\frac{\pi}{180}$ (≈ 0.0175)
$\frac{180°}{\pi}$ (≈ 57.2°)		1

3. Arc Length and Area of Sector 弧長與扇形的面積

Let 設 arc (弧) *APB* = *l*

area of sector (扇形面積) *APB* = *A*



$$\frac{l}{2\pi r} = \frac{\theta}{2\pi}$$

$$l = r\theta$$

$$\frac{A}{\pi r^2} = \frac{\theta}{2\pi}$$

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}rl$$

其中 θ 是以弧度 (radian) 為單位。

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Conversion (轉換) between Degree and Radian

Conversion of Special Angles 特別角度之轉換 :

In Degree 角度	30°	45°	60°	90°
In Radian 弧度	$\frac{\pi}{6}$	$\frac{\pi}{4}$		

In Degree 角度	120°	135°	150°	180°
In Radian 弧度				

In Degree 角度	210°	225°	240°	270°
In Radian 弧度				

In Degree 角度	300°	315°	330°	360°
In Radian 弧度				



Examples (you may use the calculator 可用計算機) :

1. $\sin\frac{\pi}{6} =$

2. $\tan\frac{\pi}{4} =$

3. $\cos\frac{\pi}{2} =$

4. $\sin\frac{3\pi}{2} =$

5. $\sec\frac{4\pi}{3} =$

6. $\cos ec\frac{7\pi}{4} =$

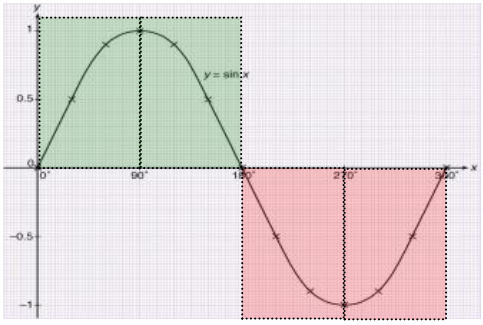
7. $\cot\frac{11\pi}{6} =$

8. $\cot 2\pi =$

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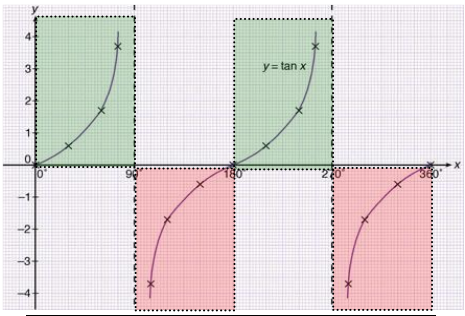
Sign（正負號）of Trigo. Ratio on $0° \le \theta \le 360°$

Graph of $y =$ sin x



+	+	−	−

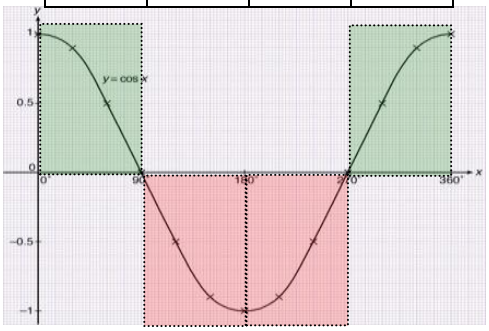
Graph of $y =$ tan x



+	−	+	−

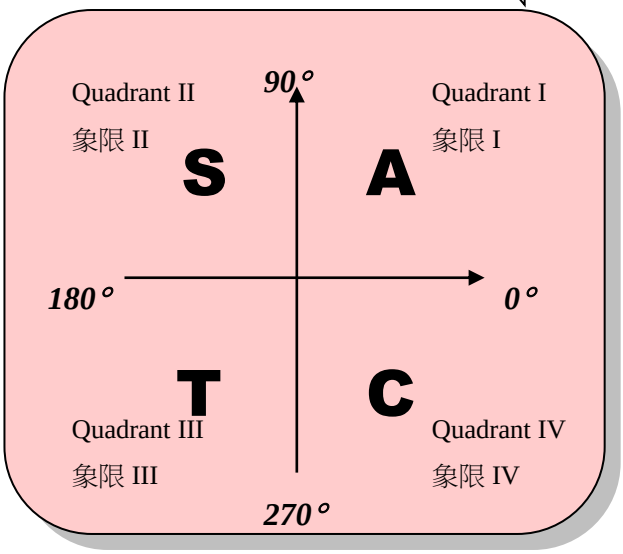
	$0° < \theta < 90°$	$90° < \theta < 180°$	$180° < \theta < 270°$	$270° < \theta < 360°$
$\sin\theta$	+	+	−	−
$\tan\theta$	+	−	+	−
$\cos\theta$	+	−	−	+

+	−	−	+



Graph of $y =$ cos x

"CAST" 圖：



Note: $\sin\theta$ 及 $\text{cosec}\theta$, $\cos\theta$ 及 $\sec\theta$, $\tan\theta$ 及 $\cot\theta$ 在相同的象限中的正負號是一樣的。

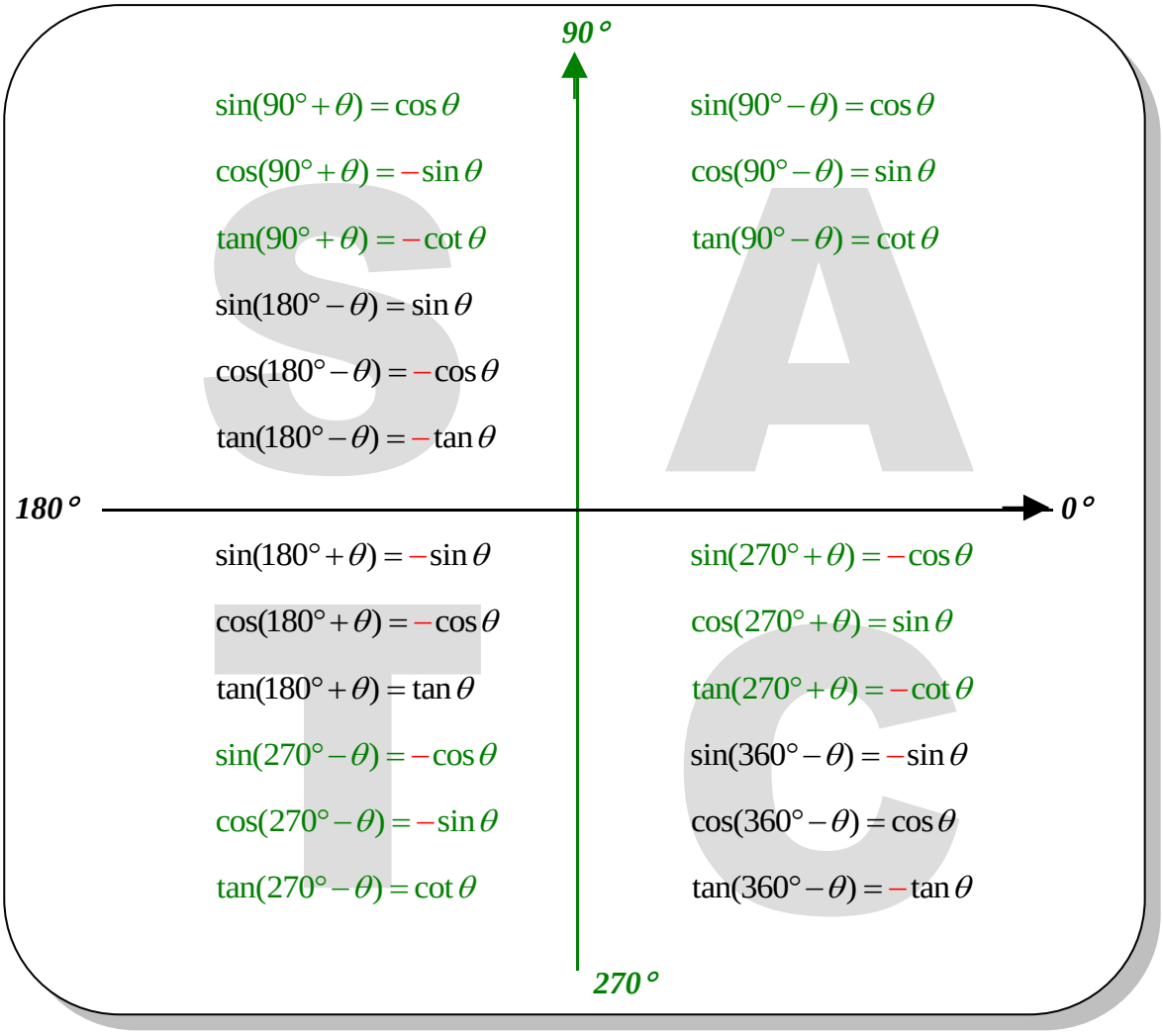
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Trigonometric Ratio（三角比）on $0° \le \theta \le 360°$



三角比極速轉換口訣！

『正負睇象限，直變橫不變！』



Examples:

cos(90° + θ)




− | ??? θ | − | sin θ

象限 II： 答案有 –ve

90° (直)： cos 變 sin！

cos(360° − θ)




+ | ??? θ | + | cos θ

象限 IV：答案是 +ve

360° (橫)：cos 不變

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The Reduction Principle (歸約法則)

1. For $0° \le \theta \le 360°$

e.g. Find the values of sin 300° without using calculator.

Sol'n

Method 1

$$\begin{aligned}&\sin 300° \\ &= \sin (270° + 30°) \\ &= -\cos 30° \\ &= -\frac{\sqrt{3}}{2}\end{aligned}$$

Method 2

$$\begin{aligned}&\sin 300° \\ &= \sin (360° - 60°) \\ &= -\sin 60° \\ &= -\frac{\sqrt{3}}{2}\end{aligned}$$



Examples (do not use calculator this time 試不用計算機 !) :

1. $\sin 120° =$ _____

$\sin 135° =$ _____

$\sin 150° =$ _____

2. $\cos 210° =$ _____

$\cos 225° =$ _____

$\cos 240° =$ _____

3. $\tan 300° =$ _____

$\tan 315° =$ _____

$\tan 330° =$ _____

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The Reduction Principle（歸約法則）

2.

$\sin(-\theta) = \sin(360° - \theta) = -\sin\theta$

$\cos(-\theta) = \cos(360° - \theta) = \cos\theta$

$\tan(-\theta) = \tan(360° - \theta) = -\tan\theta$

e.g. $\sin(-135°)$
$= -\sin 135°$
$= -\frac{\sqrt{2}}{2}$

3.

$\sin(360°n + \theta) = \sin\theta$

$\cos(360°n + \theta) = \cos\theta$

$\tan(180°n + \theta) = \tan\theta$

where ***n*** is positive integer

e.g. $\sin 390°$
$= \sin(360° + 30°)$
$= \sin 30°$
$= \frac{1}{2}$



Examples (also do not use calculator 都是不用計算機 !) :

1. $\sin 420° =$ _____

$\cos 900° =$ _____

$\tan 3660° =$ _____

2. $\sin(-60°) =$ _____

$\cos(-330)° =$ _____

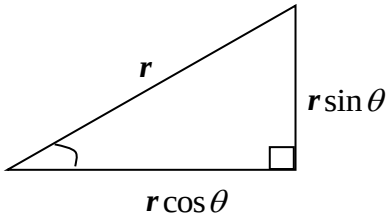
$\tan(-840°) =$ _____

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Trigonometric Identities（三角恒等式）

1. $\sin^2\theta+\cos^2\theta=1$

2. $\tan\theta=\dfrac{\sin\theta}{\cos\theta}$



將 $\sin^2\theta+\cos^2\theta=1$ 除以 $\cos^2\theta$

$\dfrac{\sin^2\theta}{\cos^2\theta}+\dfrac{\cos^2\theta}{\cos^2\theta}=\dfrac{1}{\cos^2\theta}$

3. $\tan^2\theta+1=\sec^2\theta$

4. $1+\cot^2\theta=\text{cosec}^2\theta$

將 $\sin^2\theta+\cos^2\theta=1$ 除以 $\sin^2\theta$

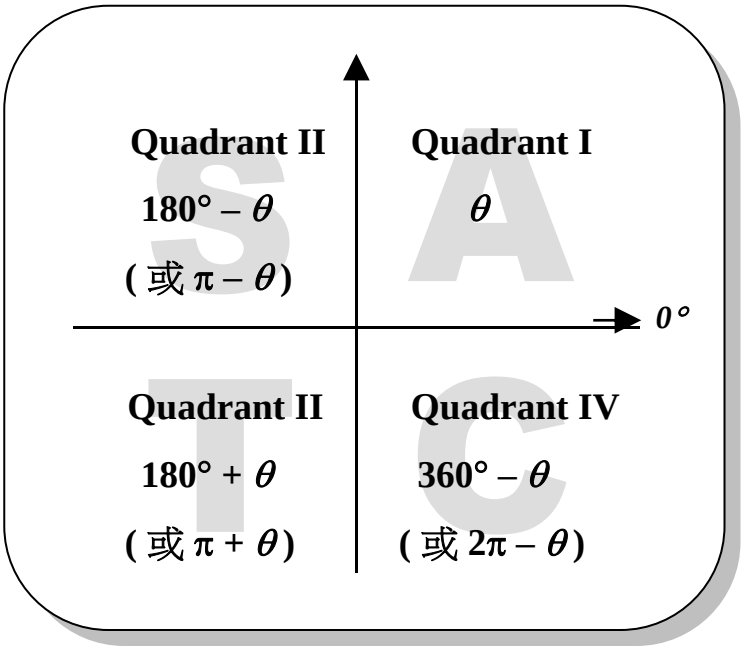
$\dfrac{\sin^2\theta}{\sin^2\theta}+\dfrac{\cos^2\theta}{\sin^2\theta}=\dfrac{1}{\sin^2\theta}$

Examples :

1. Prove $\sin\theta\sec\theta+\cos\theta\,\text{cosec}\,\theta=\sec\theta\,\text{cosec}\,\theta.$

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Trigo. Equations（三角方程）on $0° \le \theta \le 360°$



解三角方程三部曲：

1. 用恒等式 / 因式分解將方程變成基本形式：
   - $\sin\theta = k$
   - $\cos\theta = k$
   - $\tan\theta = k$
2. 篤機計 θ
   (但不要理『負』號 !!!)
3. 睇正負號知象限，
   加加減減求答案！

e.g.1 Solve $2\cos A = \sqrt{3}$ for $0° \le A \le 360°$.

$$2\cos A = \sqrt{3}$$

$$\cos A = \frac{\sqrt{3}}{2}$$

先求 θ (你可用計算機按 $(\sqrt{3} \div 2)$ SHIFT cos)

$$A = 30°, \quad 360° - 30°$$

象限 I　象限 IV

因為 cos A 是 正數
∴ A 在 象限 I 或 IV

$$A = 30°, 330°$$

e.g.2 Solve $\cot B = -\sqrt{3}$ for $0 \le B \le 2\pi$.

$$\cot B = -\sqrt{3}$$

將 cot B 變成 tan B 可令計算變得更容易.

$$\tan B = -\frac{1}{\sqrt{3}}$$

求 θ (先不理 –ve 號!!)
$(1 \div \sqrt{3})$ SHIFT tan)

$$B = \pi - \frac{\pi}{3}, \quad 2\pi - \frac{\pi}{3}$$

象限 II　象限 IV

因為 tan B 是 負數
∴ B 在 象限 II 或 IV

$$B = \frac{2\pi}{3}, \frac{5\pi}{3}$$

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Types of Trigo. Equations（三角方程的不同種類）

1. Basic Form 基本形式

e.g. Solve $2\cos A = -\sqrt{3}$ for $0 \le A \le 360°$.

Sol'n:

$$2\cos A = -\sqrt{3}$$

$$\cos A = -\frac{\sqrt{3}}{2}$$

$$A = 180° - 30°,\ 180° + 30°$$

$$A = 150°, 210°$$

2. "sin = cos" Form 形式

Tips: 盡量用 $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

e.g. Solve $\sin\theta = \sqrt{3}\cos\theta$ for $0° \le \theta \le 360°$

Sol'n:

$$\sin\theta = \sqrt{3}\cos\theta$$

整條算式除以 $\cos\theta$，
只係得番 $\tan\theta$ 就易計得多

$$\frac{\sin\theta}{\cos\theta} = \frac{\sqrt{3}\cos\theta}{\cos\theta}$$

$$\tan\theta = \sqrt{3}$$

$$\therefore \theta = 60°, 360° - 60°$$

$$\theta = 60°, 300°$$

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Types of Trigo. Equations（三角方程的不同種類）

3. Quadratic / Factor Form 形式

Tips: 請多用 $\sin^2\theta+\cos^2\theta=1$

e.g. Solve $\cos^2\theta+\sin\theta+1=0$ for $0°\le\theta\le360°$

Sol'n:

$\cos^2\theta+\sin\theta+1=0$

$(1-\sin^2\theta)+\sin\theta+1=0$ ← 將整條算式以 $\sin\theta$ 為主角 ($\cos\theta$ 或 $\tan\theta$ 都得)

$-\sin^2\theta+\sin\theta+2=0$

$(\sin\theta-2)(\sin\theta+1)=0$ ← 二次方程亦可篤機求答案

$\sin\theta=-1$ or 2 (rej.)

$\therefore\theta=270°$

4. Multiple Angles Form 複角形式

Tips: 先在 $0°\le\theta\le360n°$ 求答案，其中 n 是 θ 的倍數。

e.g. Solve $\tan2\theta=\sqrt{3}$ for $0°\le\theta\le360°$

$\therefore\ 0°\le2\theta\le720°$!

Sol'n:

$\tan2\theta=\sqrt{3}$

$2\theta=60°,\ 240°,\ 360°+60°,\ 360+240°$

$2\theta=60°,\ 240°,\ 420°,\ 600°$ ← 這題型會有很多個答案，請小心將它們全部找出！

$\therefore\theta=30°,120°,210°,300°$

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Types of Trigo. Equations（三角方程的不同種類）



Examples :

1. Solve $\sin^3 x - \cos^3 x = 0$ for $0 \le x < 2\pi$.

---

2. Solve $16\sin^4 x - 9 = 0$ for $0 \le x < 2\pi$.

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Types of Trigo. Equations（三角方程的不同種類）



Examples :

3. Solve $5\sin x - 2\cos^2 x - 1 = 0$ for $0 \le x < 2\pi$.

---

4. Solve $2\cos 3x = 1$ for $0 \le x < 2\pi$.

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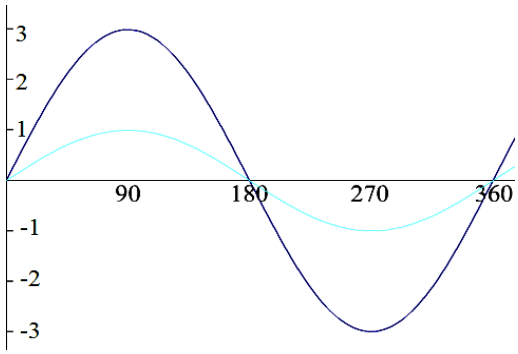
Graph of Trigo. Functions（三角函數的圖像）

$$y = \pm \mathbf{A} \sin \mathbf{B}(x \pm \mathbf{C}) \pm \mathbf{D}$$

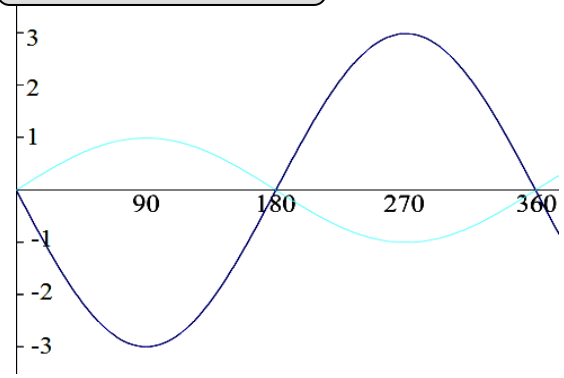
- ± : 冇事 / − : 反轉
- A : 高矮 Amplitude
- B : 肥瘦 (Frequency: No.of waveform in 360°)
- C : + : 向左走 / − : 向右走
- D : + : 向上走 / − : 向下走

e.g.

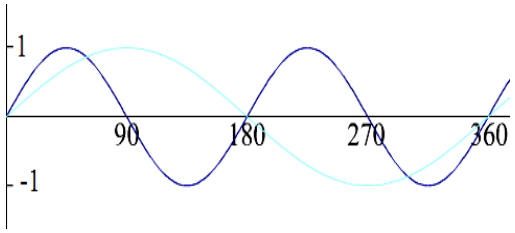
$y = 3 \sin x$



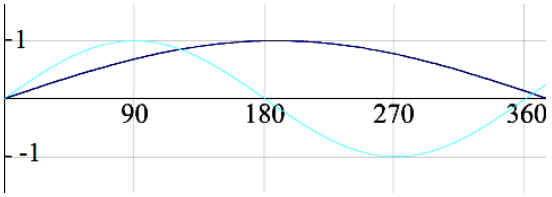
$y = -3 \sin x$



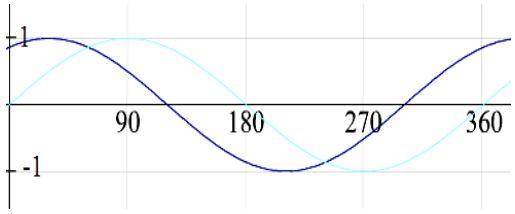
$y = \sin 2x$



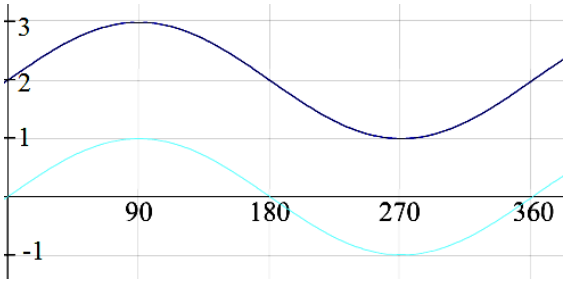
$y = \sin 0.5x$



$y = \sin (x + 30°)$



$y = \sin x + 2$



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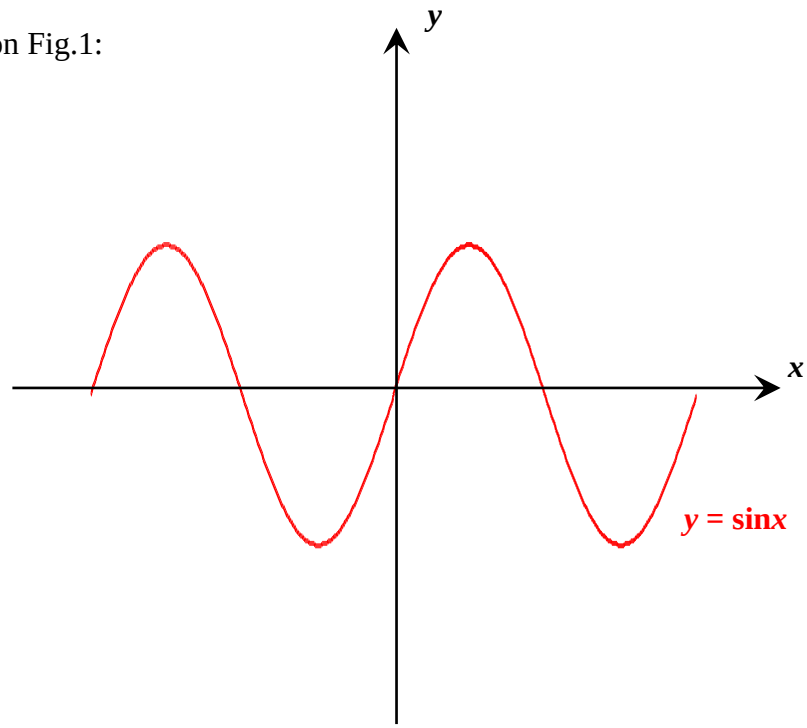
Graph of Trigo. Functions（三角函數的圖像）

God bless you ~ :

1. Sketch the follow graphs on Fig.1:

(a) $y = \sin\frac{1}{2}x$

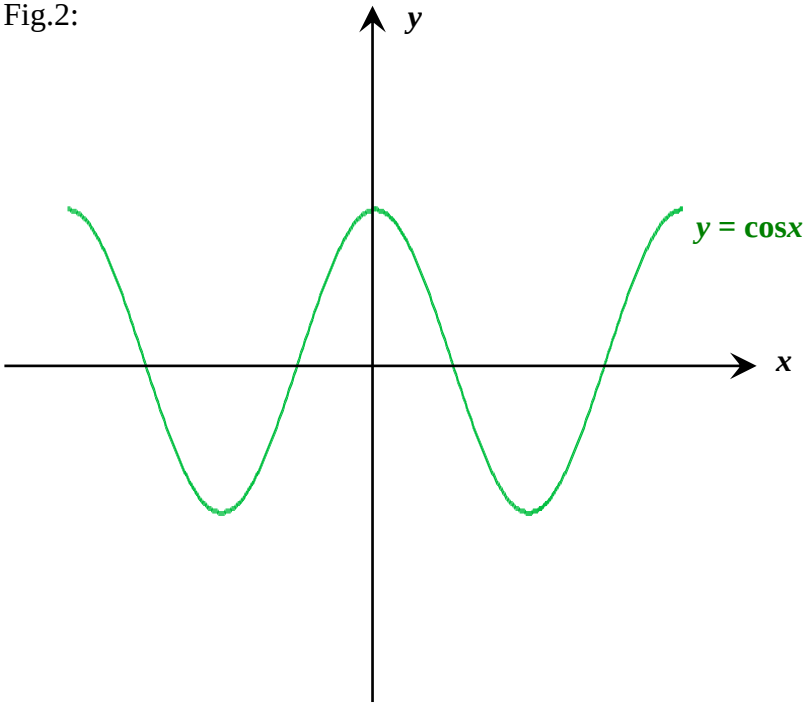
(b) $y = 2\sin\frac{1}{2}(x+\frac{\pi}{4})$



2. Sketch the follow graphs on Fig.2:

(a) $y = -\cos x$

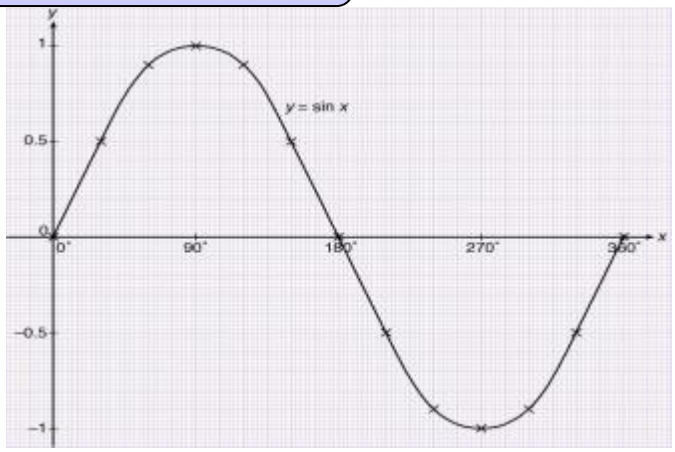
(b) $y = -\cos x - 1$



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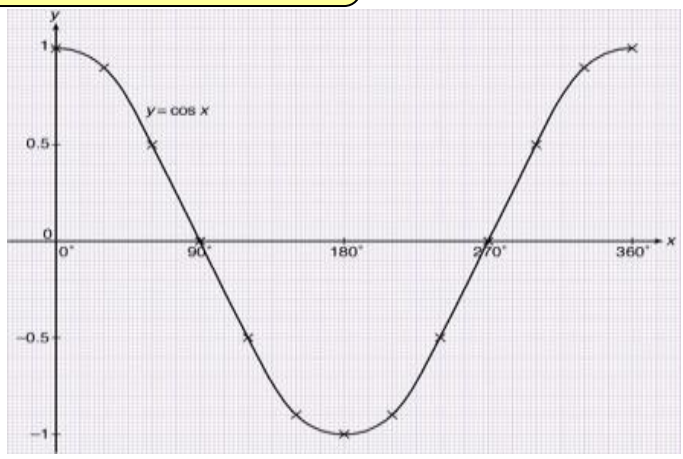
Maximum and Minimum values（最大與最小值）

Graph of $y = \sin x$



1. 週期 (Period) = 360°
2. 最大 = 1, 最小 = –1
   $-1 \le \sin x \le 1$

Graph of $y = \cos x$



1. 週期 (Period) = 360°
2. 最大 = 1, 最小 = –1
   $-1 \le \cos x \le 1$

e.g. Find the range of values of $f(x) = \dfrac{1}{\cos\theta + 2}$ for any real value of x.

$$-1 \le \cos\theta \le 1$$

$$1 \le \cos\theta + 2 \le 3$$

$$1 \ge \frac{1}{\cos\theta + 2} \ge \frac{1}{3}$$

$$\therefore \frac{1}{3} \le f(x) \le 1$$

Important!!!

For $0 < x < y \Rightarrow \dfrac{1}{x} > \dfrac{1}{y}$

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Maximum and Minimum values（最大與最小值）

Final Drill :

For any real values of x, find the range of values of the following functions:

1. $f(x) = 3\sin x - 1$

2. $f(x) = 2\cos^2 x + 1$

3. $f(x) = \dfrac{1}{2\cos x + 5}$

4. $f(x) = \sin^2 x - 2\sin x$

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